

Lesson 1: Introduction to Apps

45 minutes

Overview

Students explore and investigate what makes an app an app. They begin by looking at and discussing five different apps. Following this, students watch a video explaining the basics of how computers work. Finally students return to the apps and consider the various inputs and outputs.

Standards

Full Course Alignment

CSP Conceptual Framework

- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

Agenda

The Pre-Unit Pulse

Warm Up (5 minutes)

Preview the Unit

Activity (30 minutes)

App Exploration

Input & Output

App Investigation

Wrap Up (10 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Identify the inputs of an app
- Identify the outputs of an app
- Identify the purpose of an app

Preparation

- Preview the How Computers Work Video
- Explore the apps used in this lesson
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson**

Modifications

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 3 Slides** - Slides
▼ Make a Copy
- **Introduction to Apps** - Slides
▼ Make a Copy

For the students

- **How Computers Work - What Makes a Computer, a Computer** - Video
(Download)
- **Unit 3 Digital Journal** - Resource
▼ Make a Copy
- **Unit 3 Interactive Notebook Journal** - Resource
▼ Make a Copy

Vocabulary

- **Input** - data that are sent to a computer for processing by a program. Can come in a variety of forms, such as tactile interaction, audio, visuals, or text.
- **Output** - any data that are sent from a program to a device. Can come in a variety of forms, such as tactile interaction, audio, visuals, or text.
- **User Interface** - the inputs and outputs that allow a user to interact with a piece of software. User interfaces can include a variety of forms such as buttons, menus, images, text, and graphics.

Teaching Guide

The Pre-Unit Pulse

Have your students complete these questions independently some time before starting this unit.

☰ 1-3

The Pre-Unit Pulse

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💡 Teaching Tip

Pre-Unit Pulse Questions: The answers to these questions can provide insights into the preferences, strengths, and motivations of your students. If you'd like to adapt the questions or add in your own, you may want to make a copy of [this Google Form](#) which already has the existing questions populated.

For more tips and ideas of how to use these questions, check out the [CSP Guide to Pre-Unit Pulse Questions](#)

Warm Up (5 minutes)

Preview the Unit

Distribute: Give each student a copy of the Unit 3 Journal ([Digital/Interactive Notebook](#)) or have students set up this unit's section in their CSP journals.

💡 Teaching Tip

Journaling: Journaling can take many different forms, but in general, it's a tool for individual processing and reflection in a form that can be revisited as students develop their skills and understandings. The medium used for journaling can vary depending on classroom needs.

Remarks

Now that we've learned how to represent information digitally and how the Internet works, it's time to think about creating digital content for others to interact with.

 **Discuss:** *What are apps? How do we interact with them? What kinds of things do apps do?*

Discussion Goal: Apps are designed to solve a problem by providing a service

- Entertainment: images, movies, tv shows, etc.
- Social: social media, chatting
- Commerce: marketplaces
- Finding Information: search engines, maps

Apps have screens or *User Interfaces* to display and collect information to and from users

- These things collect information: Buttons, sliders, text boxes, etc.
- This is the type of information that can be displayed: Sounds, images, video, text

Activity (30 minutes)

App Exploration

Remarks

Today we are going to be looking at some sample apps to explore their purpose and function.

Teaching Tip

This lesson is an introduction to a unit that introduces programming through the broader context of designing apps that help people. This lesson is designed to build excitement about the skills that students will develop throughout the unit and get them thinking early about the ways programming can help others. This lesson also establishes shared vocabulary for talking about apps in a generic sense, for example its inputs, outputs, and overall user interface.

 **Group:** Organize students in groups of two. Students should navigate to the App Exploration starting on level 1.

Levels 4-8: There are five different sample apps. Depending on time, students can explore 3-5 different apps.

App #1: Water Conservation Tips

- The user clicks through the different screens and makes choices about the correct ways to conserve water. When the user makes choices, images or sounds are played to indicate if a choice is the right one.
- The purpose of the app is to teach people about ways to conserve water.
- The target audience is young people who are interested in the topic

App #2: Bird Quiz

- The user answers trivia questions about birds. If the answer is correct, they go to the next question. If wrong, they are taken to a losing screen, and then start the quiz again. If three questions are answered correctly the user is taken to the winning screen.
- The purpose of the app is to teach users different facts about birds
- The target audience is anyone interested in birds

App #3: Hamilton Township

- The user clicks through a few different screens to learn about township efforts to beautify the town
- The purpose of the app is to educate users on steps the town is taking to improve the town
- The target audience is people who live in Hamilton Township

App #4: Four Square Instructions

- The user navigates through different screens that explain how to play four square. On one screen, the user is quizzed on how many people are needed to play four square.
- The purpose of the app is to instruct the user on the rules of four square.
- The target audience is people who would like to play four square.

App #5: Monarch Butterflies

- The user clicks on different screens and pictures of monarch butterflies to explore the life stages of the butterfly
- The purpose of the app is to educate users on the life cycle of the monarch butterfly
- The target audience is young people, most likely children or teenagers who are interested in life science and butterflies



4-8

App Exploration

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 **Discuss:** *With a partner, discuss the following and note down in your journal:*

- *How does the user interact with the app?*
- *What is the overall purpose of this app?*
- *Who is the target audience?*

Share Out: As a class, discuss student answers to the discussion questions.



Teaching Tip



Use the **Retrieve-Pair-Share** strategy to facilitate this discussion by directing students to write down their thoughts and ideas first, then pair up students to compare their answers and expand their notes. This allows time for individual processing and idea generation before sharing thoughts with others. Ask students to share the ideas they discussed with their partners as a class.

For more ideas on how to adapt this strategy to fit the needs of your class, check out the [CSP Guide to Classroom Discussions](#)



Remarks

In developing apps, which are a kind of computing innovation, it's important to understand both their purpose and function. The purpose helps the developer focus on creating an app with specific goals in mind.

Now let's dig in to how the user interacts with the apps.

Input & Output

 **Display:** How Computers Work [How Computers Work - What Makes a Computer, a Computer](#)



Teaching Tip



Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

Remarks

The video explained *Input, Storage, Processing, and Output*. Today we are going to focus on input and output. An app on a phone gets input in many different ways from a user or from another program and displays (or in the case of sound: plays) output. The output of a program is usually based on inputs or the initial way the program was set up. For example, you could program an app to play a song as soon as it runs.

App Investigation

 **Levels 9-13:** With your partner, take another look at the sample apps you explored before by navigating to the App Investigations starting at level 6. Consider what the inputs and outputs are for the apps. Note these down in your journal.

Teaching Tip

After students are finished writing in their journals, discuss as a class or collect the journals to review student answers.

- An input could be a user clicking a button or tapping the screen
- An output could be an image displayed or a sound played



9-13

App Investigation

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Wrap Up (10 minutes)

Remarks

Apps have unique purposes that appeal to different types of users. Apps take in input from users and output information in various ways. Users interact with apps through the *user interface*.

 **Discuss or Write:** Focus on an app from this level and explain its purpose. Give an example of who a potential user of such an app might be.

Depending on the amount of time that you have available in class, you can choose to either have a whole-class discussion of the prompt or a writing opportunity for students to practice their skills for the new Create PT Task.



Assessment Opportunity

Create Performance Task Written Response Practice

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Sample Prompt: *Focus on an app from this level and explain its purpose. Give an example of who a potential user of such an app might be.*

Unit Journal Template

Sample Response for Instructors (coming soon...)

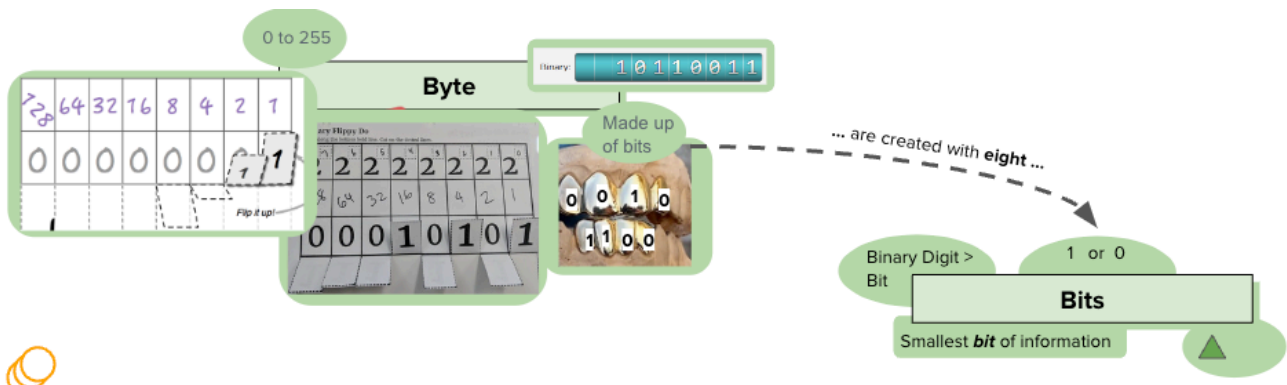
Discussion / Writing Goal: Have students share their thoughts with a partner. Then, bring the class back together and discuss a few examples.

 **Journal** Add to your journal the following terms along with your own definitions:

- User Interface
- Input
- Output

Teaching Tip

Vocabulary Canvas: You may choose to add your own examples to the slides. The image below shows an example of a Vocabulary Canvas. Also, you can find definitions and graphics for the vocabulary words in the hidden slides at the end of the lesson slides if you'd like to use those with your students.



Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Match term with the example.



Check For Understanding



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